Data Base System Lab 3

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INLAB TASKS

1.

Worksheet | Query Builder

```
CREATE TABLE Employees (
    Employee_ID   NUMBER PRIMARY KEY,
    Employee_Name VARCHAR2(100) NOT NULL,
    Hire_Date     DATE,
    Job_Title     VARCHAR2(50),
    Salary        NUMBER(10,2)
);
```

Script Output | Query Result

Task completed in 0.024 seconds

```
Table EMPLOYEES created.
```

2.

```
CREATE TABLE Departments (
    Department_ID   NUMBER PRIMARY KEY,
    Department_Name VARCHAR2(100) UNIQUE NOT NULL
);
```

Script Output | Query Result

Task completed in 0.04 seconds

```
Table EMPLOYEES created.

Table DEPARTMENTS created.
```

3.

Worksheet | Query Builder

```
ALTER TABLE Employees
ADD Department_ID NUMBER;
```

Script Output | Query Result

Task completed in 0.333 seconds

```
Table EMPLOYEES created.

Table DEPARTMENTS created.

Table EMPLOYEES altered.
```

4.

Welcome Page | abcd

Worksheet | Query Builder

```
ALTER TABLE Employees
ADD CONSTRAINT fk_emp_dept
FOREIGN KEY (Department_ID)
REFERENCES Departments(Department_ID);
```

```
Table EMPLOYEES created.

Table DEPARTMENTS created.

Table EMPLOYEES altered.

Table EMPLOYEES altered.
```

5.

Worksheet | Query Builder

```
CREATE TABLE Students (
    Student_ID   NUMBER PRIMARY KEY,
    Student_Name VARCHAR2(100),
    Age          NUMBER CHECK (Age >= 18),
    Gender       VARCHAR2(10)
);
```

```
Table STUDENTS created.
```

6.

```
CREATE TABLE Products (
    Product_ID   NUMBER PRIMARY KEY,
    Product_Name VARCHAR2(100),
    Price        NUMBER(10,2) CHECK (Price > 0)
);
```

Script Output × | Query Result ×

Task completed in 0.068 seconds

```
Table PRODUCTS created.
```

7.

Worksheet | Query Builder

```
CREATE TABLE Users (
    User_ID  NUMBER PRIMARY KEY,
    Username VARCHAR2(50),
    Email    VARCHAR2(100) UNIQUE
);
```

Script Output × | Query Result ×

Task completed in 0.027 seconds

```
Table USERS created.
```

8.

Worksheet | Query Builder

```
CREATE TABLE Orders (
    Order_ID    NUMBER PRIMARY KEY,
    Order_Date  DATE,
    Status      VARCHAR2(20) DEFAULT 'Pending'
);
```

Script Output | Query Result

Task completed in 0.026 seconds

```
Table USERS created.

Table ORDERS created.
```

9.

```
INSERT INTO Departments (Department_ID, Department_Name)
VALUES (1, 'Human Resources');
```

Script Output | Query Result

Task completed in 0.016 seconds

```
Table USERS created.

Table ORDERS created.

1 row inserted.
```

10.

Worksheet | Query Builder

```
INSERT ALL
    INTO Employees (Employee_ID, Employee_Name, Hire_Date, Job_Title, Salary, Department_ID)
    VALUES (101, 'Ali', DATE '2022-01-15', 'CEO', 600000, 1)
    INTO Employees (Employee_ID, Employee_Name, Hire_Date, Job_Title, Salary, Department_ID)
    VALUES (102, 'Murtuza', DATE '2023-03-10', 'Analyst', 455000, 1)
    INTO Employees (Employee_ID, Employee_Name, Hire_Date, Job_Title, Salary, Department_ID)
    VALUES (103, 'Sayyam', DATE '2021-07-20', 'Clerk', 30000, 1)
SELECT * FROM dual;
```

Script Output | Query Result

Task completed in 0.022 seconds

```
3 rows inserted.
```

POSTLAB TASK

11.

Worksheet | Query Builder

```
ALTER TABLE Employees
MODIFY Job_Title VARCHAR2(50) NOT NULL;
```

12.

```
ALTER TABLE Students
ADD Status VARCHAR2(20);

ALTER TABLE Students
MODIFY Status VARCHAR2(20) DEFAULT 'Active';
```

Script Output | Query Result

Task completed in 0.044 seconds

```
Table STUDENTS altered.

Table STUDENTS altered.
```

13.

Worksheet | Query Builder

```
ALTER TABLE Employees
RENAME TO Staff;
```

14.

```
ALTER TABLE Students
RENAME COLUMN Student_Name TO Full_Name;
```

Script Output | Query Result

Task completed in 0.041 seconds

```
Table STUDENTS altered.
```

15.

Worksheet | Query Builder

```
CREATE TABLE Payments (
    Payment_ID NUMBER PRIMARY KEY,
    Amount NUMBER(10,2) NOT NULL,
    Payment_Date DATE
);
```

Script Output × | Query Result ×

Task completed in 0.036 seconds

```
Table PAYMENTS created.
```

16.

```
CREATE TABLE Courses (
    Course_ID NUMBER PRIMARY KEY,
    Course_Name VARCHAR2(100)
);
```

Script Output × | Query Result ×

Task completed in 0.044 seconds

```
Table COURSES created.
```

17.

```
CREATE TABLE Enrollments (
    Enrollment_ID NUMBER PRIMARY KEY,
    Student_ID NUMBER,
    Course_ID NUMBER,
    CONSTRAINT fk_enroll_student
        FOREIGN KEY (Student_ID)
        REFERENCES Students(Student_ID),
    CONSTRAINT fk_enroll_course
        FOREIGN KEY (Course_ID)
        REFERENCES Courses(Course_ID)
);
```

Script Output × Query Result ×

Task completed in 0.044 seconds

```
Table ENROLLMENTS created.
```

18.

```
ALTER TABLE Products
ADD CONSTRAINT uk_product_name UNIQUE (Product_Name);
```

Script Output × Query Result ×

Task completed in 0.041 seconds

```
Table PRODUCTS altered.
```

19.

```
CREATE TABLE Teachers (
    Teacher_ID NUMBER PRIMARY KEY,
    Teacher_Name VARCHAR2(100),
    Experience NUMBER CHECK (Experience >= 1)
);
```

Script Output × | Query Result ×

Task completed in 0.042 seconds

```
Table PRODUCTS altered.

Table TEACHERS created.
```

20.

Worksheet | Query Builder

```
INSERT INTO Orders (Order_ID, Order_Date)
VALUES (1001, DATE '2024-01-15');

SELECT * FROM Orders WHERE Order_ID = 1001;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 1 in 0.012 seconds

	ORDER_ID	ORDER_DATE	STATUS
1	1001	15-JAN-24	Pending